Welcome to the amos2025ss04-ai-driven-testing wiki!

Our project

How to use

User and Developer Guide (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/User-and-Developer-Guide) For a quick start and for the correct setup. You may also want to check out specific sections for certain applications:

- How to start the Webinterface Frontend (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/How-to-start-the-Webinterface---Frontend) Tutorial on how to start it
- Ollama Setup Information (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Ollama-Setup-Information) Possible ways to use Ollama to run LLMs
- Possible Ways to run Ollama and Settings (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Possible-Ways-to-run-Ollama-and-Settings)

How to work on this project

- Contributing (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Contributing)
- How to be a Release Manager (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/How-to-be-a-Release-Manager) How to do the weekly release. This is relevant or any future team tat takes part in the AMOS course.

Structure

- Architecture (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Architecture) Overview of the system architecture and component interactions.
- Backend API Design (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Backend-API-Design) The structure recieving prompts, interacting with LLMs and returning the results to the client.
- CI Pipeline (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/CI-Pipeline-Overview) CI/CD Pipeline Overview over the continuous integration pipeline for the project.
- Execution Flow Design (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Execution-Flow-Design) TODO: finish this entry
- Model Configuration Design (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Model-Configuration-Design) How we handel models.
- Module System Design (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Module-System-Design) How we handel the extentions.
- Prompt Design (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Prompt-Design) Some Guidance on effective prompting.

- Robot Framework (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Robot-Framework) Information about potential integration with the Robot Framework.

Models

A list of all language models evaluated for use in the project. We only chose the versions of models that are avaiable under an open source license.

- DeepCoder (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/DeepCoder)
- DeepSeek-Coder V1 (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/DeepSeek%E2%80%90Coder-V1)
- Google Gemma 3 (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Google-Gemma-3)
- LLMs incompatibility with our project (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/LLMs-incompatibility-with-our-project) Overview of models that were evaluated but deemed unsuitable.
- Mistral AI (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Mistral-AI)
- OpenHermes 2.5 (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/OpenHermes-2.5)
- Phi4-Mini (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Phi4%E2%80%90Mini)
- Phi-4 Reasoning (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Phi%E2%80%914-Reasoning)
- Qwen 2.5 Coder (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Qwen-2.5-Coder)
- Qwen3 (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Qwen3)
- Smollm2 (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Smollm2)
- StarCoder (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/StarCoder)
- StarCoder2 (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/StarCoder2)
- TinyLlama (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/TinyLlama)

Our research and experiments

- Docker Performance (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Docker-Performance) Performance of different docker configurations when running LLMs
- Evaluating Large Language Model Responses to Spelling Errors (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Evaluating-Large-Language-Model-Responses-to-Spelling-Errors)
- LLM Code Understanding Evaluation (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/LLM-Code-Understanding-Evaluation)
- Comparison of 1b, 3b, 7b and 14b models (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Comparison-of-1b,-3b,-7b-and-14b-models)
- Running AI LLM Projects in CI (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Running-AI-LLM-Projects-in-CI)

LLM components and assessment tools

- AI-Model-Benchmark (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/AI%E2%80%90Model-Benchmark) Standard Benchmarks used to evaluate LLMs
- Benefits of chaining LLMs (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Benefits-of-chaining-LLMs)
- Code Complexity (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Code-Complexity) Description of the code complexity metrics that will be used for evaluation, nicluding MCC and CCC.
- Code Coverage (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Code-Coverage)
- Include a project as context (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Include-a-project-as-context) This explains how to include repositories into the LLM promt as context using a RAG and the alteratives.
- Iterative Refinement (Multi-Pass Generation) (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Iterative-Refinement-(Multi%E2%80%90Pass-Generation)) Passing the responce as the new prompt input.

Training an LLM by yourself

- How to train a LLM (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/How-to-train-a-LLM)
- How to run on the FAU HPC (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/How-to-run-on-the-FAU-HPC)
- Choosing the Right Dataset for LLM Training on the University HPC (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Choosing-the-Right-Dataset-for-LLM-Training-on-the-University-HPC)
- How to finetune a LLM (https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/How-to-finetune-a-LLM)

The maintenance of this Wiki officially stops at 16.07.2025, as this is the Demo-day of our project and no further contribution is expected from the team.